If α is not an integer, φ^α is only defined where $\varphi > 0$. Since $x - x_0$ and $\frac{1}{x-x_0}$ are negative for $x < x_0$, and x and $\frac{1}{x}$ are negative as $x \to -\infty$, in these cases we use $|\varphi|$ instead of φ (for example $|x - x_0|^\alpha$).

Order of an infinitesimal/infinite function

Let f be infinitesimal or infinite at x_0. If there exists $\alpha > 0$ such that

$$f \asymp \varphi^\alpha, \quad x \to x_0$$

then we say that α is the **order** of f at x_0 with respect to φ.

Observe that if f has an order it is *unique*. Furthermore, by definition of the symbol $\asymp$, $f \asymp \varphi^\alpha$, $x \to x_0$, means that

$$\lim_{x \to x_0} \frac{f(x)}{\varphi^\alpha(x)} = \ell \in \mathbb{R} \setminus \{0\}.$$

This can also be written as:

$$f \sim \ell\varphi^\alpha, \quad x \to x_0$$

which can be rewritten as

$$f = \ell\varphi^\alpha + o(\varphi^\alpha), \quad x \to x_0.$$

Principal part of an infinitesimal/infinite function

If $f = \ell\varphi^\alpha + o(\varphi^\alpha)$ as $x \to x_0$ then

$$p(x) = \ell\varphi^\alpha(x)$$

is called the **principal part** of f at x_0 with respect to φ.

Examples

Example 8.4: Consider the function

$$f(x) = \sin x - \tan x, \qquad \text{near } x_0 = 0.$$

Since $\sin 0 = \tan 0 = 0$, f is infinitesimal at $x = 0$. We can write:

$$\begin{aligned} \sin x - \tan x &= \sin x - \frac{\sin x}{\cos x} \\ &= \frac{\sin x \cdot (\cos x - 1)}{\cos x} \\ &\sim \frac{x \cdot (-\frac{1}{2}x^2)}{1} \\ &= -\frac{1}{2}x^3, \qquad x \to 0. \end{aligned}$$

Hence f is infinitesimal of order 3 at $x = 0$ with respect to $\varphi(x) = x$. The principal part is $p(x) = -\frac{1}{2}x^3$ and we can write:

$$\sin x - \tan x = -\frac{1}{2}x^3 + o(x^3), \quad x \to 0.$$

Example 8.5: Consider the function

$$f(x) = \sqrt{x^2+3} - \sqrt{x^2-1}.$$

As $x \to +\infty$, $f(x)$ tends to 0 (verify that you can see why!). So f is infinitesimal at $x_0 = +\infty$. We'll want to compare it to $\varphi(x) = \frac{1}{x}$. First we want the roots in the denominator, so we multiply and divide by $\sqrt{x^2+3} + \sqrt{x^2-1}$ to get:

$$\begin{aligned} f(x) &= \sqrt{x^2+3} - \sqrt{x^2-1} \\ &= \frac{(x^2+3)-(x^2-1)}{\sqrt{x^2+3}+\sqrt{x^2-1}} \\ &= \frac{1}{x} \cdot \frac{4}{\sqrt{1+\frac{3}{x^2}}+\sqrt{1-\frac{1}{x^2}}} \end{aligned}$$

To compare to $\varphi(x) = \frac{1}{x}$ we need to look at

$$\lim_{x\to+\infty} \frac{f(x)}{\varphi^\alpha(x)}$$

and try to identify the correct α. We see that we get:

$$\lim_{x\to+\infty} \frac{f(x)}{\varphi^\alpha(x)} = \lim_{x\to+\infty} \frac{x^\alpha}{x} \cdot \frac{4}{\sqrt{1+\frac{3}{x^2}}+\sqrt{1-\frac{1}{x^2}}}$$

so that with $\alpha = 1$ we have

$$\lim_{x\to+\infty} \frac{f(x)}{\varphi^\alpha(x)} = \lim_{x\to+\infty} \frac{4}{\sqrt{1+\frac{3}{x^2}}+\sqrt{1-\frac{1}{x^2}}} = 2.$$

Hence f is infinitesimal of order 1 at $+\infty$ with respect to $\varphi(x) = x^{-1}$. The principal part is $p(x) = 2x^{-1}$ and we can write

$$\sqrt{x^2+3} - \sqrt{x^2-1} = 2x^{-1} + o(x^{-1}), \qquad x \to +\infty.$$

Example 8.6: Consider the function

$$f(x) = \sqrt{9x^5+7x^3-1}$$

which is infinite as $x \to +\infty$. To determine the order we compare it with $\varphi(x) = x$:

$$\begin{aligned} \lim_{x\to+\infty} \frac{f(x)}{x^\alpha} &= \lim_{x\to+\infty} \frac{\sqrt{9x^5+7x^3-1}}{x^\alpha} \\ &= \lim_{x\to+\infty} \frac{x^{\frac{5}{2}}\sqrt{9+7x^{-2}-x^{-5}}}{x^\alpha}. \end{aligned}$$

This suggests choosing $\alpha = \frac{5}{2}$. Then we have:

$$\lim_{x\to+\infty} \frac{f(x)}{x^{\frac{5}{2}}} = \lim_{x\to+\infty} \sqrt{9 + 7x^{-2} - x^{-5}} = 3.$$

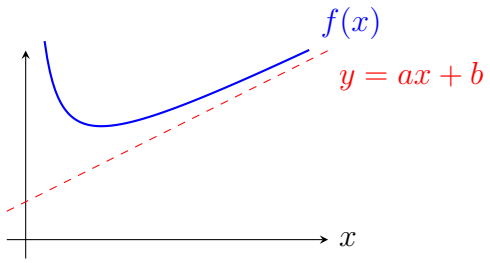
Hence f is infinite of order $\frac{5}{2}$ at $+\infty$ with respect to $\varphi(x) = x$. The principal part is $p(x) = 3x^{\frac{5}{2}}$ and we can write

$$\sqrt{9x^5 + 7x^3 - 1} = 3x^{\frac{5}{2}} + o(x^{\frac{5}{2}}), \qquad x \to +\infty.$$

8.4 Asymptotic behavior

We have already seen *horizontal* and *vertical* asymptotes. However it is possible to have slanted asymptotes. We say that a function $f(x)$ **behaves asymptotically as** $x \to +\infty$ like the affine function $y = ax + b$ if

$$\lim_{x\to+\infty} (f(x) - (ax + b)) = 0.$$



A similar definition can be made for $x \to -\infty$.

If $a \neq 0$, the line $y = ax + b$ is called an **oblique asymptote** of f at $+\infty$ (if $a = 0$ it is a horizontal asymptote). In the language of this chapter, $f(x) = ax + b + o(1)$ as $x \to +\infty$.

Proposition 8.6: The line $y = ax + b$ is an asymptote of f at $+\infty$ if and only if

$$a = \lim_{x\to+\infty} \frac{f(x)}{x} \in \mathbb{R} \qquad \text{and} \qquad b = \lim_{x\to+\infty} \big(f(x) - ax\big) \in \mathbb{R}.$$

Proof. If $f(x) - ax - b \to 0$, then $f(x) - ax \to b$ and $\frac{f(x)}{x} = a + \frac{f(x)-ax}{x} \to a$. Conversely, if both limits exist, then $f(x) - (ax + b) = (f(x) - ax) - b \to 0$. □

Example 8.7: Let $f(x) = \sqrt{x^2 + x}$. Then $\frac{f(x)}{x} = \sqrt{1 + \frac{1}{x}} \to 1$ as $x \to +\infty$, so $a = 1$, and

$$f(x) - x = \frac{x^2 + x - x^2}{\sqrt{x^2 + x} + x} = \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} \to \frac{1}{2},$$

so $y = x + \frac{1}{2}$ is an oblique asymptote of f at $+\infty$. (As $x \to -\infty$ one finds $y = -x - \frac{1}{2}$ instead.)

In principle we can compare a function $f : \mathbb{R} \to \mathbb{R}$ to any function $g : \mathbb{R} \to \mathbb{R}$ as $x \to \pm\infty$. In the case of $+\infty$, we say that f behaves asymptotically as $x \to +\infty$ like g if

$$\lim_{x\to+\infty} (f(x) - g(x)) = 0.$$

Chapter 9

Global properties of continuous functions

Up to now we studied *local* properties of functions (that is, properties of functions as $x \to x_0$, where x_0 can be a finite point or $\pm\infty$). In the current chapter we study *global* properties of functions. That is, we consider functions defined on intervals $[a,b] \subset \mathbb{R}$ (where $a < b$) and analyze their behavior on this entire interval.

Continuity on a closed interval

Throughout this chapter, "f is continuous on $[a,b]$" means that f is continuous at every point of (a,b), right-continuous at a and left-continuous at b.

Theorem 6.15(7a) still holds at the endpoints for sequences in $[a,b]$: if $x_n \in [a,b]$ for all n and $\lim_{n\to\infty} x_n = a$, then $\lim_{n\to\infty} f(x_n) = f(a)$, and similarly at b (only points of $[a,b]$ are involved, so one-sided continuity suffices).

9.1 Existence of zeros: Bolzano's Theorem

A **zero** of a function f is a point x_0 such that $f(x_0) = 0$. Our first result guarantees the existence of zeros:

Theorem 9.1 (Bolzano's Theorem)**:** Let $f : \mathrm{dom}(f) \subseteq \mathbb{R} \to \mathbb{R}$ and suppose it is continuous on a *closed* interval $[a,b] \subseteq \mathrm{dom}(f)$ where $a < b$. If $f(a)$ and $f(b)$ have opposite signs, i.e. $f(a)f(b) < 0$, then there exists $x_0 \in (a,b)$ such that $f(x_0) = 0$. Moreover, if f is *strictly* monotone on $[a,b]$ then x_0 is the *unique* point in $[a,b]$ where f equals zero.

Proof. Assume that $f(a) < 0 < f(b)$. The proof for the case $f(a) > 0 > f(b)$ will follow the exact same ideas. We split the proof into several steps. The ideas are essentially identical to the *bisection method* that we've seen in the proof of the Bolzano-Weierstrass Theorem (see §5.3)

Step A. Constructing sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$ in search of x_0:

Step A0. Define $a_0 = a$ and $b_0 = b$. Define the center point between a_0 and b_0 to be $c_0 = \frac{a_0+b_0}{2}$. If $f(c_0) = 0$ then $x_0 = c_0$ and the proof is done. Otherwise, if $f(c_0) < 0$ we define $a_1 = c_0$ and $b_1 = b_0$ and if $f(c_0) > 0$ we define $a_1 = a_0$ and $b_1 = c_0$.

Step A1. Define the center point between a_1 and b_1 to be $c_1 = \frac{a_1+b_1}{2}$. If $f(c_1) = 0$ then $x_0 = c_1$ and the proof is done. Otherwise, if $f(c_1) < 0$ we define $a_2 = c_1$ and $b_2 = b_1$ and if $f(c_1) > 0$ we define $a_2 = a_1$ and $b_2 = c_1$.

Step A2. Define the center point between a_2 and b_2 to be $c_2 = \frac{a_2+b_2}{2}$. If $f(c_2) = 0$ then $x_0 = c_2$ and the proof is done. Otherwise, if $f(c_2) < 0$ we define $a_3 = c_2$ and $b_3 = b_2$ and if $f(c_2) > 0$ we define $a_3 = a_2$ and $b_3 = c_2$.

...and so on...

Step An. The points a_n and b_n have been defined in the previous step. Define the center point between a_n and b_n to be $c_n = \frac{a_n+b_n}{2}$. If $f(c_n) = 0$ then $x_0 = c_n$ and the proof is done. Otherwise, if $f(c_n) < 0$ we define $a_{n+1} = c_n$ and $b_{n+1} = b_n$ and if $f(c_n) > 0$ we define $a_{n+1} = a_n$ and $b_{n+1} = c_n$.

...and so on...

Step B. Properties of the sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$.

If a zero x_0 hasn't been located in any finite step, then by induction we have constructed sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$ satisfying:

$$\begin{aligned}
&a_0 \le a_1 \le \cdots \le a_n \le \cdots \le b_n \le \cdots \le b_1 \le b_0\\
&[a_0, b_0] \supset [a_1, b_1] \supset \cdots \supset [a_n, b_n] \supset \cdots\\
&f(a_n) < 0 < f(b_n) \quad \text{and} \quad b_n - a_n = \frac{b_0 - a_0}{2^n}, \quad \forall n \in \mathbb{N}.
\end{aligned}$$

By the first property, $\{a_n\}_{n\in\mathbb{N}}$ is a bounded, monotonically increasing sequence (it is bounded by b_0, for instance). Hence, by the Monotone Convergence Theorem (Theorem 5.1), its limit exists; denote it by $a_\infty \in \mathbb{R}$:

$$\lim_{n\to\infty} a_n = a_\infty.$$

Similarly, $\{b_n\}_{n\in\mathbb{N}}$ is bounded and monotonically decreasing, hence its limit exists by the same theorem. We denote it by $b_\infty \in \mathbb{R}$:

$$\lim_{n\to\infty} b_n = b_\infty.$$

Step C. Conclusion.

Observe that

$$b_\infty - a_\infty = \lim_{n\to\infty} b_n - \lim_{n\to\infty} a_n = \lim_{n\to\infty} (b_n - a_n) = \lim_{n\to\infty} \frac{b_0 - a_0}{2^n} = 0$$

so that $b_\infty = a_\infty$. Denote this number by x_0:

$$x_0 = b_\infty = a_\infty.$$

Since $a_n \in [a_0, b_0] = [a, b]$ for all n, Proposition 5.3 gives $x_0 \in [a, b]$. Since f is continuous, we may use Theorem 6.15(7a) (the substitution theorem for a continuous function of a convergent sequence) to get:

$$\lim_{n\to\infty} f(a_n) = \lim_{n\to\infty} f(b_n) = f(x_0).$$

However we know that

$$f(a_n) < 0 < f(b_n), \quad \forall n \in \mathbb{N}.$$

By Theorem 6.15(4),

$$f(x_0) = \lim_{n\to\infty} f(a_n) \leq 0 \quad \text{and} \quad 0 \leq \lim_{n\to\infty} f(b_n) = f(x_0)$$

Hence it necessarily follows that

$$f(x_0) = 0.$$

Since $f(a) < 0 < f(b)$, $x_0 \neq a$ and $x_0 \neq b$, hence $x_0 \in (a, b)$.

Finally, if f is strictly monotone on $[a, b]$, then by Proposition 4.1, f is injective on $[a, b]$. This means that for every $y \in f([a, b])$ there exists a unique $x \in [a, b]$ such that $f(x) = y$. Hence x_0 is the unique zero of f in $[a, b]$. □

> **The word "algorithm"**
>
> *The proof above is also a practical way of computing a zero of f, known as the* bisection method *(we already used the same idea in the proof of the Bolzano-Weierstrass Theorem, Section 5.3, keeping each time the half that contains infinitely many terms of the sequence). After n steps we know that x_0 lies in $[a_n, b_n]$, an interval of length $\frac{b-a}{2^n}$, so the midpoint c_n approximates x_0 with an error of at most $\frac{b-a}{2^{n+1}}$. Every step halves the error: starting from an interval of length 1, twenty steps bring it below 10^{-6} (since $2^{20} > 10^6$). Other methods are faster, but bisection cannot fail as long as f is continuous and changes sign, and it is still used in numerical software, often combined with faster methods.*
>
> *A step-by-step procedure of this kind is called an* algorithm. *The word comes from Muhammad ibn Musa al-Khwarizmi, a scholar at the House of Wisdom in Baghdad, the academy founded in the 9th century by Caliph al-Ma'mun, where Greek scientific works were translated into Arabic. Al-Khwarizmi wrote a treatise explaining the decimal place-value system developed in India (where Brahmagupta had already given rules for computing with zero in the 7th century). Its Latin translation opens with* Dixit Algoritmi *("al-Khwarizmi says");* algorismus *came to mean computing with these numerals, and was later respelled "algorithm" by confusion with the Greek* arithmos *("number"). His book on solving equations,* al-jabr*, gave us "algebra." This is also why the digits we use are called "Arabic numerals" in Europe.*

Example 9.1: The function

$$f(x) = \begin{cases} 1 & x \geq 10 \\ -1 & x < 10 \end{cases}$$

changes sign on the interval $[9, 11]$. However, there is no zero in $[9, 11]$ (i.e. there is no $x \in [9, 11]$ such that $f(x) = 0$). Why? The problem with this f is that it is not continuous on $[9, 11]$ (there's a jump discontinuity at $x = 10$).

Example 9.2: The function $f(x) = x^2$ is a continuous function on $\mathbb{R}$ that has a zero at $x = 0$, however it is always non-negative (i.e. it does not change sign). This shows that a continuous function can have a zero without changing sign. That is, changing sign is a *sufficient* condition for a continuous function to have a zero, but not a *necessary* condition.